

Muhammad Faizan Anjum Shah

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Computer Systems Engineer specializing in ML pipeline development, DSP-based feature engineering, and hardware-software integration. Experienced in building and deploying end-to-end ML systems from data ingestion through real-time inference on both embedded devices and cloud-connected architectures. Formal DSP training from EPFL (Coursera) combined with academic coursework provides a strong analytical foundation for signal classification, prediction, and optimization. Certified in ML and Deep Learning by DeepLearning.AI (Andrew Ng, 2025).

EDUCATION

B.E. in Computer Systems Engineering

Expected June 2026

Sukkur IBA University, Pakistan — GPA 3.60/4.00, Merit Scholar

- Concentrations: Intelligence & Modelling, Computer Vision, Digital Signal Processing, Embedded Systems
- Relevant Coursework: Machine Learning, Deep Learning, AI, Digital Signal Processing, Probability & Statistics, Robotics & Automation, Embedded Systems, Database Systems

CERTIFICATIONS

- Deep Learning Specialization (5 courses: DNNs, CNNs, Sequence Models, ML Strategy, NLP) — DeepLearning.AI / Andrew Ng (2025)
- Machine Learning Specialization — DeepLearning.AI / Andrew Ng (2024)
- Digital Signal Processing — EPFL / Coursera (signal theory, filtering, spectral analysis, DTFT, z-transforms)
- MATLAB & Simulink for Engineers — Coursera (model-based design, simulation, algorithm prototyping)
- Python — Kaggle (data manipulation, pipelines, applied ML)

EXPERIENCE

Data Visualization & ML Pipeline Associate Intern

Jul – Aug 2025

Excelerate — Remote

- Ingested, cleaned, and validated 1M+ rows across six heterogeneous datasets; engineered a schema-reconciled master table used directly for downstream reporting and analysis.
- Built interactive KPI dashboards in Looker Studio with drill-down filters and automated chart refresh.
- Applied completeness, range, and consistency checks before aggregation.

PROJECTS

AI-Powered Lead Intelligence & Automation System — Python · Flask · Groq Llama 3.3 70B · n8n · SQLite · REST API

- Designed an end-to-end ML data pipeline with CSV ingestion, DNS/MX email validation, LLM-based feature extraction (industry, company size, lead score 1–10, action recommendation), SQLite persistence, and a live Flask dashboard.
- Built a classification and scoring pipeline on unstructured data using Groq Llama 3.3 70B as a structured feature extractor.
- Orchestrated n8n workflows with weekly scheduled enrichment, conditional branching on hot_leads thresholds, and webhook intake from external systems.
- Developed a 10-endpoint REST API with async enrichment, live progress polling, and CSV export

Vision Transformer for Lung Disease Classification — PyTorch · ViT · Grad-CAM · Medical Imaging

- Trained a Vision Transformer model using the transfer learning on chest X-ray datasets, achieving 91.27% accuracy across the full train/validation/test pipeline with confusion matrix and per-class precision/recall evaluation.
- Applied Grad-CAM attention visualization for model interpretability to help understand the areas that results in the actual prediction.
- Executed the complete ML lifecycle including data preprocessing, augmentation, hyperparameter tuning, evaluation, and thorough documentation of results.

Urdu Speech Emotion Detection & Audio Enhancement — Python · MFCC · Wavelet Transform · DSP · Classification

- ▶ Implemented MFCC feature extraction from raw Urdu speech audio, applied wavelet-based denoising, and trained an emotion classifier — delivering a full signal processing pipeline from acquisition to labeled output.
- ▶ Enhanced audio using frequency-domain filtering.

IoT Anomaly Detection (Isolation Forest on Sensor Streams) — Python · Scikit-learn · Isolation Forest · Real-Time Sensor Data

- ▶ Built an unsupervised anomaly detection pipeline for continuous IoT sensor streams, tuning the contamination parameter against labelled fault events and evaluating precision/recall.
- ▶ Mapped directly to ML-based classification and prediction tasks in engineering platforms, including sensor signal analysis, fault detection, and real-time inference loops.

PSL Cricket Win Probability Engine — XGBoost · LightGBM · PyTorch · Time-Aware Ensemble · Simulation

- ▶ Hybrid ensemble (XGBoost, LightGBM, PyTorch) with time-aware win probability simulation; cricket-domain feature engineering including run rate deltas, team strength, toss decision, venue, and head-to-head.
- ▶ Demonstrated sequential and temporal data handling, ensemble design, probability calibration, and model validation.

Smart Glove (Embedded ML Gesture Translator) — Arduino · C++ · Sensor Fusion · Real-Time Inference

- ▶ Developed an end-to-end embedded system combining flex sensors and IMU with a C++ inference loop to classify gestures with sub-50ms latency on Arduino.
- ▶ Achieved full hardware-software integration from sensor acquisition to model output, leveraging UART communication and interrupt-driven I/O for ML deployment at the edge.

Real-Time Gesture Recognition Pipeline — Python · MediaPipe · OpenCV · Inference Optimisation

- ▶ Implemented a MediaPipe-based gesture detection pipeline to ~30 FPS on CPU, converting hand gestures into words through latency budgeting, profiling, and bottleneck elimination.

TECHNICAL SKILLS

ML / DL	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM; CNNs, Vision Transformers (ViT), Ensemble Methods, Anomaly Detection
Signal Processing	MFCC, Wavelet Transform, FFT, Spectral Analysis, Frequency-domain Feature Engineering, Filtering (EPFL DSP certified)
Data Engineering	Pandas, NumPy, SQL/SQLite, PostgreSQL, Multi-source Pipeline Design, Feature Engineering, SMOTE, Schema Validation
System Architecture	Flask REST APIs , n8n Automation, Async Pipelines, Webhook Integration, Groq LLM API, SQLite
Embedded / HW-SW	Arduino, ESP32, Raspberry Pi, C/C++, Real-Time Inference, UART/SPI/I2C, Sensor Fusion
MATLAB / Simulink	Model-based Design, System Simulation, Algorithm Prototyping (Coursera certified)
Programming	Python (primary), C/C++, MATLAB, SQL
Dev & Viz Tools	Git, Jupyter, Looker Studio, GAZEBO, ROS

LEADERSHIP & COMMUNITY

- ▶ Co-Trainer, Generative AI Workshop — Served as co-trainer at a Generative AI workshop, assisting learners with LLM fine-tuning, LoRA/QLoRA optimization, few-shot prompting, and diffusion-based image generation using varied parameters.
- ▶ Active Member, Data Science Society — Sukkur IBA University; participated in peer-led ML projects, workshops, and knowledge-sharing initiatives.
- ▶ Participant, See Pakistan Startup Pitching — pitched a data-driven waste management solution; demonstrates ability to translate technical ML work into real-world product thinking.